

CAMPBREAK · REQUIREMENTS

CampBreak requirements

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CBK-REQ-001	Version	0.3
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/campbreak		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3 with the decisions of CBK-DDR-001 and CBK-DDR-002 applied and results from CBK-CAL-001
0.2	2026-10-03	Amish Chadha	Populated to TRL 2 with concept media, components and first-order numbers
0.1	2026-09-30	Amish Chadha	Initial scaffold

CampBreak requirements

Measurable requirements for one block kit, with the TRL 3 result from the calculation note CBK-CAL-001. Nine are met on paper or by design, R2 can only be shown in a cooking trial at TRL 4, and **R9 is not met on paper** for a shelter 100 m from the station.

Table 1. Requirements and TRL 3 status.

ID	Requirement	Target	Verification (TRL 4 or later)	TRL 3 result	Status
R1	Alarm detects a fast temperature rise	Alarms at a rise of 8 °C (14 °F) per minute or faster, plus a 57 °C fixed threshold	Heat chamber test with logged temperature ramp	Threshold set in firmware, averaged over 60 s	Met by design
R2	Alarm ignores normal cooking	No alarm during 1 hour of stove cooking at 1 m distance	Cooking trial in a mock shelter	Not calculable on paper	To show at TRL 4
R3	Alarm is loud enough to wake the block	At least 85 dB(A) at 3 m	Sound level meter test	85.5 dB(A)	Met (est.)
R4	Alarm reaches neighbours and the block station	Every alarm in the block and the station within 10 s	Field test across a mock block layout	7.2 s worst case	Met (est.)
R5	Alarm runs without mains power	At least 12 months on batteries	Current draw measurement	Limited by cell shelf life, about 5 years	Met (est.)
R6	Cart can be moved on camp paths	Two adults move the loaded cart 100 m on an unpaved 10 % slope in 3 minutes or less; cart width 0.9 m or less	Timed trial on a test track	1.7 min; 89 N per person; 850 mm wide	Met (est.)
R7	Hand pump delivers useful flow	At least 20 L/min (5.3 US gal/min) at the nozzle, pumped by two people for 10 minutes	Bench flow test with timed volume	20.4 L/min at 47 W per person	Met (est.)
R8	Jet reaches a burning shelter from a safe distance	Jet reach of 6 m (20 ft) or more	Measured throw test	6.9 m	Met (est.)
R9	Fast deployment	Water on target within 3 minutes of alarm for a shelter 100 m from the station	Timed drill with volunteers	3.8 min at 100 m; 2.8 min at 70 m with a 30 s muster	Not met on paper
R10	Repairable locally	All wear parts replaceable with hand tools and parts sold in local markets	Maintenance walkthrough with a camp workshop	Every wear part is a market item	Met by design
R11	Cost against the value-engineering target	Value-engineering target: USD 1,600 for one block kit	Costed bill of materials	USD 1,355, USD 245 under the target	Met

R9 stays at 100 m. The decided mitigation (CBK-DDR-001, D9) is a siting rule: no shelter more than 70 m along the path from a station, with a drilled 30 s muster.

■ Assumptions

- A block has a drum or tap within reach of most shelters, or drums are placed at stations.
- Camp management allows alarms in shelters and a station in each block.
- Volunteers are trained and drill regularly, so a 30 s muster is realistic.
- The estimates marked (est.) in CBK-CAL-001 hold within the margins shown; the pump displacement is the most important one.

■ Safety

Safety: These requirements describe a pressurised hand-pumped water system and battery-powered alarms. The kit is for small, early fires only, people leave first, and water is never used on burning oil or live wiring. It is an open engineering reference, not certified equipment.